

Gain and Abstraction

This idea that there is generality in the specific is of far-reaching importance.

—DOUGLAS R. HOFSTADTER¹

An abstraction is taking a point of view or looking at things under a certain aspect or from a particular angle. All sciences are differentiated by their abstraction.

—FULTON J. SHEEN²

Love is one kind of abstraction. And then there are those nights when I sleep alone, when I curl into a pillow that isn't you, when I hear the tiptoe sounds that aren't yours. It's not as if I can conjure you up completely. I must embrace the idea of you instead.

—DAVID LEVITHAN³

Mounted archery was used for hunting and warfare by most nomads of the Eurasian steppe including the Huns, Magyars, and Mongols, as well as by Native Americans on the American plains. Later, as warfare and hunting came to depend on guns, archery evolved into a martial art. Imagine trying to shoot an enemy warrior, a moving target, while your horse speeds along at 50 kilometers an hour. Perhaps nobody knows more about shooting arrows from a galloping horse than Lajos Kassai, known as “The

1. Hofstadter (1979).

2. <https://www.diannasnape.com/abstraction/>.

3. Levithan (2011).

Master” of this ancient art in the remote Hungarian village where my ancestors originated. While on his horse, Kassai appears super-human; he never misses a target.⁴ He says masters have an instinct for aiming—after decades of arduous training. Mounted archers rely on a combination of eye–hand coordination, muscle activity patterns, arrow trajectory, and subconscious distance calculations to hit the target. There are just too many things going on to allow deliberate calculation when you aim to hit 10 targets in 20 s. The archer’s eyes, head, and torso; the body on the horse; and the horse’s body, head, and eyes are all embedded in their own independent coordinate frames. Multiply all this by the same number of degrees of freedom for the enemy warrior and then calculate his angle and direction of movement.

Riding a horse, moving your eyes, or exploring the environment exploits the same principles of coordination. Every coordinate is a different viewpoint (Chapter 10). In every coordinate frame, the information is first generalized and then segregated into categories by some new observer-centered classifier. To achieve effective motor–sensory action, downstream reader networks should extract information according to their points of view.⁵ Gain is a mechanism that assists with such coordinated transformation and abstraction problems.

PRINCIPLES AND MECHANISMS OF GAIN

Turning the volume knob on your radio is a gain control mechanism for sound production.⁶ Gain modulation requires two sources, an amplifier and a modulator (Figure 11.1). When the sources are multiplied (or divided), such interaction generates an output whose magnitude is larger or smaller (in which case the gain is negative) than expected from adding (or subtracting) the sources.

4. You can watch him in action here: <https://www.youtube.com/watch?v=a0opKAKbyJw>; <https://www.youtube.com/watch?v=2yorHswhzrU>. Kassai trained Hollywood actor Matt Damon in how to handle the bow in the movie *The Great Wall*.

5. Artificial intelligence algorithms also face the problem of hierarchical abstraction. The best performing deep Q network (DQN; Minh et al., 2017) exploits the local correlations present in images and makes its way up through transformations of viewpoints and scales. The program’s explicit goal is to select actions in a fashion that maximizes reward. Thus, grounding is based on (imitated) actions.

6. In electrical engineering, gain is often used synonymously with amplification or boosting the signal. Gain of an amplifier is measured in decibels (dB), a logarithmic unit. A tenfold input–output increase corresponds to a gain of 10 dB; a 100-fold increase represents 20 dB (or 10 to the power of 1 and 2, respectively). The power spectrum of the local field potential (LFP) and electroencephalogram (EEG) is often characterized by such gain factors.

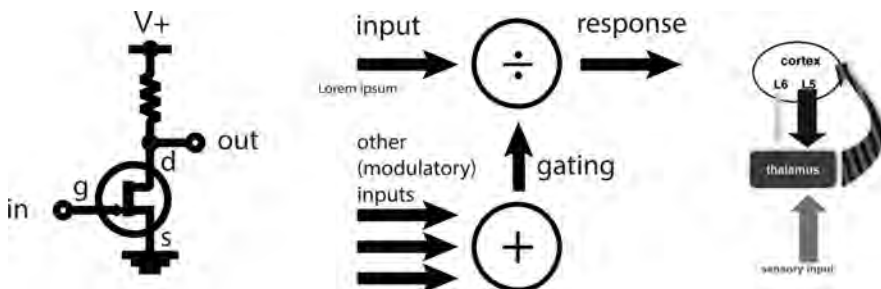


Figure 11.1. *Left:* Gain control in electronic circuits. The magnitude of current flowing from the source (s) to the drain (d) in a semiconductor can be adjusted by applying a small current to its gate (g). Small changes in the input (gate) result in large changes in the output voltage (V_{out}). *Middle:* General implementation of gain. Input–output transfer is modulated by a division or multiplication circuit, modulated by a second, gating input. All subcortical neuromodulators, such as norepinephrine and acetylcholine, achieve their gain control on neurotransmission by this principle. *Right:* Cortical gain control of sensory input transmission in the thalamus. Thalamocortical neurons receive convergent inputs originating from subcortical (gray arrow) and cortical (from layer 5 and layer 6) sources. Layer 6 input rarely discharges thalamocortical neurons, but it can boost their response to sensory inputs, thus providing gain control. Layer 5 inputs can effectively discharge thalamocortical cells and can contribute to the maintenance of persistent thalamocortical activity that can outlast the input. The output from the thalamus is represented by the curved black and gray striped arrow. Thalamic figure from Groh et al. (2014).

Such nonlinear gain modulation is ubiquitous in the brain, with far-reaching consequences, and is achieved by different mechanisms in individual neurons or synapses or at a larger scale in neuronal networks, indicating that it is a critical operation. Gain control can also determine whether the succession of events in a neuronal trajectory is slow and fast. Thus, gain-controlled rate of change in neuronal sequences can, in principle, replace the concept of time (Chapter 10). Here, I list a few ways in which gain control is implemented in neuronal circuits. If you are not interested in the details, you are welcome to skip the next few subsections.

Neuronal Implementation of Gain Control

The intensity of light hitting the retina varies over nine orders of magnitude (a billion-fold change) from a dark night to a sunny day. Many peripheral mechanisms, including changes in pupil diameter and chemical adaptation of the cones and rods in the retina, effectively reduce the light responses of retinal

ganglion cells and the messages sent to the visual system. Still, visual neurons have to deal with a wide dynamic range. The solution is to respond not to the absolute magnitude of the input but to a relative change in intensity, known as *contrast*. By receiving information about both the mean background intensity and about the deviation from this mean, neurons can calculate the ratio or contrast against any light intensity. This process is a type of normalization, to borrow a term from statistics—essentially a division.⁷

Shunting

Multiple mechanisms contribute to achieving such a wide sensitivity range. The first mechanism is gain control in an individual neuron, a change in that neuron's response to a given input based on its other inputs. Each neuron receives hundreds to thousands of synapses from excitatory and inhibitory neurons. Whether it responds with an output action potential depends on the ratio of excitation to inhibition across all its inputs. But it also matters whether that ratio comes from 5 excitatory and 2 inhibitory neurons or from 500 excitatory and 200 inhibitory neurons because neurotransmitters work by opening channels, essentially making temporary holes in the membrane. When many channels are open, the resistance between the extracellular and intracellular environments decreases. In other words, the conductance through the membrane increases as ions can flow freely between the inside and the outside of the neuron. The technical term for this state is “shunting” because the numerous open channels work as a short circuit. This is similar to the situation that occurs when you turn on the oven, toaster, and dishwasher at the same time and the circuit breaker pops because the parallel resistances shunt the circuit. Therefore, even if the membrane voltage remains the same, as happens when the increases in excitation and inhibition are balanced, the inputs have less impact on the neuron when many channels are open because the increased conductance (a short-circuit) reduces the efficacy of the synaptic current. In essence, the efficacy of any synaptic input to the neuron is divided, or normalized, by the combined effects of all other inputs impinging on the neuron at the same time.⁸ A change in gain alters the neuron's sensitivity to changes in its inputs

7. Normalization is a special case of gain modulation, where the “normalizer” (denominator) is a superset of the signals that determine the responses (Carandini and Heeger, 1994, 2012). In this case, input from one neuron is the factor to be affected by the numerous other inputs to the same neuron.

8. This normalization effect is a consequence of Ohm's law, as the synaptic input current I at a given membrane potential V is scaled by membrane conductance g , as $I = gV$. Multiplication by

without affecting its selectivity or discriminative properties. The gain or efficacy of synapses is amplified or discounted by the background activity of the neuronal population in a multiplicative (divisive), as opposed to additive (subtractive) manner.⁹

Inhibition

A second mechanism of gain control occurs in inhibitory synapses acting through gamma aminobutyric acid (GABA_A) receptors.¹⁰ These receptors are permeable to chloride ions, which have an equilibrium potential close to the resting membrane potential of the neuron. As a result, activation of GABA_A receptors barely changes the membrane potential of a relatively quiet neuron, but it shunts the neuron and thus brings about a negative gain (division) on excitatory inputs.¹¹ This mechanism is important for the sustainability of cell assemblies because they compete with each other through lateral inhibition. Such gain control mechanisms affect the transmission of spikes from neuron to neuron by affecting spike generation.

Short-Term Plasticity

A third family of gain control mechanisms acts on the inputs to the neurons. When many nearby inputs on a dendrite are activated, they amplify each other's

a given constant changes the slope, or gain, of the input–output relationship, corresponding to an up- or down-scaling of responsiveness.

9. An increase in neuronal gain corresponds to a multiplication, and a decrease corresponds to a division. A nice summary of neuronal arithmetic is by Silver (2010).

10. Alger and Nicoll (1979); Kaila (1994); Vida et al. (2006).

11. For more details, see http://www.scholarpedia.org/article/Neural_inhibition. Shunting synapses are more effective if the inhibitory synapse is on the path between an excitatory synapse and the spike initiation site. Shunting is spatially localized, whether it is caused by inhibition or excitation because both inhibitory postsynaptic current and excitatory postsynaptic current potential cause leaks in the membrane by opening channels. Gain modulation by shunting becomes more apparent when large numbers of neurons are affected and when there is a relatively balanced increase of excitation and inhibition in short time windows, as typically occurs during gamma frequency oscillations. Balanced background activity produces an approximately multiplicative gain modulation of the response to an excitatory input (Chance et al., 2002).

effects so that their impact is larger than their algebraic sum.¹² Even more selective gain control, acting at single synapses, can be achieved via the short-term plasticity of synaptic strength. The synaptic efficacy of the action potential from a presynaptic neuron is not fixed over time but shows large fluctuations. Some synapses increase (“facilitatory synapses”) whereas others decrease (“depressing synapses”) their response to the later input spikes in a series.¹³ Depression is the predominant form of short-term plasticity in synapses between cortical pyramidal neurons. When the input neuron fires slowly, the magnitude of the evoked postsynaptic responses is similar. However, when the neuron fires quickly, the magnitude of the postsynaptic response diminishes over time. The faster the intervals between input spikes, the stronger the synaptic depression. After the barrage of spikes is over, recovery of the baseline response may take about a second, with a characteristic time constant for each synapse type. Because depressing synapses become increasingly less efficient with increasing rates of input spikes, they can be conceived as low-pass filters of neuronal transmission. Conversely, facilitating synapses are high-pass filters. As a result, the response of the postsynaptic neuron will reflect not the average input rate of the presynaptic neuron but the change in its rate. Short-term synaptic plasticity, therefore, turns neurons into “ratio meters.” The nonlinearity of the short-term plasticity transforms the input–output relationship via divisive gain changes.¹⁴ Thus, while the mechanisms are different, gain control operates on both single synapses and single neurons. Gain control through dynamic changes of the synapse can adjust the rate of change across neurons in neuronal sequence, which can be conceptualized as time warping (Chapter 10).

Neuromodulation

In neuronal networks, gain control is achieved by mechanisms that affect large numbers of similar neurons. Recurrent excitation is a frequent source of

12. Rodolfo Llinás was among the first to demonstrate that dendrites are not simply passive integrators but have active voltage-sensitive channels that can powerfully enhance the neuron's computation ability (Llinás, 1988). See also Mel (1999); Magee (2000).

13. Frequency-dependent short-term potentiation (STP) occurs when presynaptic vesicle supplies are quickly depleted or when postsynaptic receptors desensitize. Short-term facilitation can also have both pre- and postsynaptic mechanisms (Thomson and Deuchars, 1994; Markram and Tsodyks, 1996; Zucker and Regehr, 2002). English et al. (2017) described various forms of spike transmission mechanisms between pyramidal cells and interneurons that can perform frequency filtering and amplification at various time scales. These flexible mechanisms can assist in shifting cell assemblies without external influence.

14. Abbott et al. (1997); Rothman et al. (2009).

nonlinear multiplication in neuronal networks. Subcortical neuromodulators, such as acetylcholine, norepinephrine, serotonin, dopamine, and histamine, share one thing in common: they affect the excitability of their target neurons to a large extent by affecting conductance, exerting multiplicative gain control on the output firing rates. In addition to excitability, neuromodulation can affect the balance between synaptic properties and nonlinear membrane properties by targeting ion channels. Several neuromodulators also target pre-synaptic terminals. Presynaptic receptors can adjust the amount of neurotransmitter to be released into the synaptic cleft, thus affecting synaptic efficacy¹⁵ Because subcortical neuromodulators typically affect many neurons simultaneously, they are critical for gain control in neuronal networks.¹⁶

Conductance changes, GABA_A receptor-mediated shunting, lateral inhibition, depressing synapses, dendritic boosting, recurrent excitation, and subcortical neuromodulation are ubiquitous features in cortical circuits. Therefore, any of these mechanisms in isolation or combination produce gain control in neuronal networks, influencing their input–output transformation in various ways. Since neuromodulators can also affect the speed of spike transmission across neuronal assemblies, such effects may explain why neuromodulators and drugs that can affect their physiological effects can lead to acceleration or slowing of subjective time.

INPUT MAGNITUDE NORMALIZATION

Perhaps the most obvious example of gain modulation is the response of sensory systems to natural stimuli of differing intensities, such as the adaptation to light intensity changes mentioned earlier. Pyramidal neurons in the primary visual cortex, the first cortical processing station that handles inputs from the thalamus, have a limited dynamic range in their firing rates, which confine their ability to respond differentially to a limited range of contrast. However, neurons that receive thalamic inputs also receive inputs from many other cortical neurons. This arrangement allows the impact of the thalamic input to be

15. Ion-channel modulation typically involves binding of a neuromodulator to G-protein-coupled receptors and subsequent activation of second messengers, although ion channels can be modulated through other mechanisms as well. The family of two-pore domain potassium (K⁺) channels are responsible for a substantial portion of leak conductance in a variety of neurons, and subcortical neurotransmitters effectively modulate leak conductance (Kaczmarek and Levitan, 1986; Bucher and Marder, 2013). This can explain their strong impact on network dynamics.

16. The role of norepinephrine in gain control is reviewed extensively by Aston-Jones and Cohen (2005).

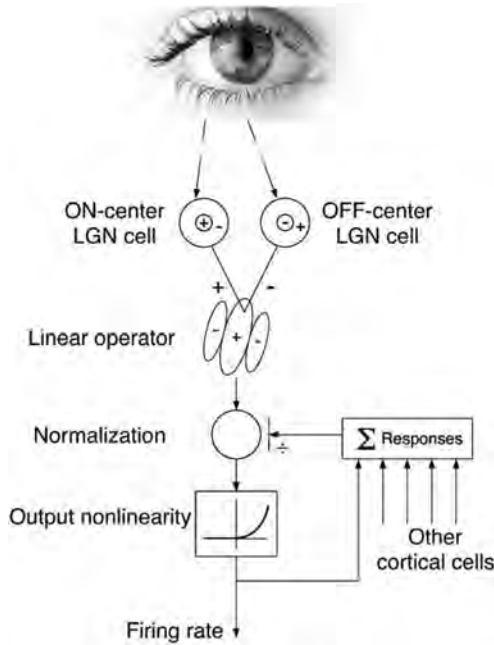


Figure 11.2. Illustration of neuronal normalization. A linear stage combines complementary inputs from the retina by way of the lateral geniculate nucleus (LGN). Neurons in the visual cortex sum responses of “on-center” neurons (i.e., those neurons that respond to the visual input) and subtracts responses of “off-center” neurons (i.e., those neurons whose receptive fields are outside those of the on-center neurons). This is a linear operation. The flanking inhibitory subregions are obtained by the opposite arrangement of excitation and inhibition. A shunting-based normalization mechanism divides the linear stage’s response by the pooled activity of a large number of cortical cells. This is a nonlinear operation. Reproduced from Carandini and Heeger (1994).

reduced proportionally (scaled down) by the conductance increase brought about by feedback excitation from many other neurons (Figure 11.2). It is essentially a division.

A related mechanism is inhibition-mediated competition. A neuron’s response in the primary visual cortex to a preferred orientation stimulus can be suppressed by superimposing a nonpreferred orientation. This happens because the activity of neurons with nonpreferred orientation excites inhibitory interneurons, which in turn suppress the neuron’s response to its own preferred stimuli.¹⁷ The concentrations of odors and sound intensities also vary

17. While there is an agreement about the principle (the need for normalization), a debate persists about whether the division is brought about by conductance change, inhibition, or other

considerably, and a similar mechanism may be responsible for computing intensity change. All it takes is two inputs, a specific one that activates the neuron and another one that produces divisive normalization.¹⁸

CHANGING VIEWS BY TRANSFORMING COORDINATE SYSTEMS

Now let us examine how gain control mechanisms can be exploited for transforming signals. When you are driving a car, your eyes are fixed on the traffic ahead. If you want to reduce the volume on the radio, you simply reach for the volume knob. In effect, you are multiplexing, attending to the traffic with central vision and to the volume knob with peripheral vision. Because human eyes move a lot in their sockets, the relative positions of the eye, head, and trunk vary from moment to moment. Thus, the retinal output and retinotopic representation of eye-centered space by neurons in the visual system cannot be of much help in the reach movement of the hand. The retinal image must be first translated into head- and body-centered coordinate frames.¹⁹

Receptive Fields and Gain Fields

Neurons in the visual system have *receptive fields*, meaning that they respond maximally when a stimulus is presented at a particular location (peak response) corresponding to a given spot on the retina (Figure 11.3).

mechanisms (Carandini and Heeger, 1994, 2011; Murphy and Miller, 2003). Normalization is believed to be at work in the retina as well, where the output spiking of bipolar and ganglion cells is divided by the activity of other ganglion cells induced by the light contrast in the surrounding regions (Normann and Perlman, 1979).

18. Competitive normalization has been described in the fly antennal lobe during olfactory processing. Here, projection neurons receive two different inputs: excitation from the olfactory receptors and inhibition from other projection neurons (called *feedforward* or *lateral inhibition*). When a single odorant is mixed with many other odorants, the response of a projection neuron to its favored odorant can be decreased by simultaneous activation of many other partner neurons by a mixture of odorants. The divisive normalization scales with the total magnitude of activity of the olfactory receptor neuron population. Increasing the activity of projection neurons makes the responses to the favorite stimuli not only weaker but also more transient (Olsen et al., 2010).

19. The mechanism that compensates for the ever-shifting retinal input relative to the body represents a corollary discharge (see Chapter 3).

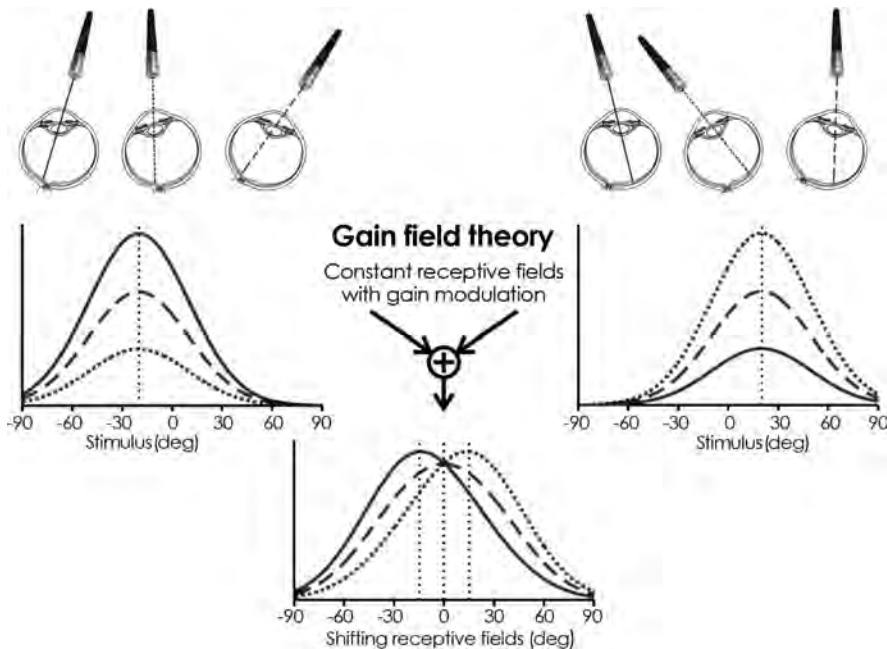


Figure 11.3. Gain field mechanisms in the parietal cortex. The upper panels show the hypothetical receptive fields of two neurons that are gain-modulated by eye position. Three lines in each graph represent visual receptive fields mapped relative to gaze at three different eye positions. The laser light can scan the entire retina. Here only one spot (its maximum receptive field response) is shown. Note that eye position modulates the strength of response of two neurons, but their receptive fields remain the same. However, summation of these two gain-modulated neural responses (*bottom panel*) results in shifting receptive fields in the output as a function of eye position. From Zipser and Andersen (1988).

The receptive field can be easily mapped when the eyes are voluntarily fixed on a target by projecting a lighted spot successively onto different positions across the visual field. The peak response and the decreasing responses that surround the peak form a cone-shaped response pattern, which is the receptive field.²⁰

20. The idea is that the visual receptive field is related to the place field of a hippocampal place cell (Chapter 6). The term “receptive field” is central to the study of sensory systems. In addition to position, visual neurons also respond to various features of the light stimulus, such as its velocity, direction, and orientation. The magnitude of responses within the receptive field can be affected by a number of factors, such as brain state. A solid observation in all fields of neuroscience is that brain state changes across sleep and waking to exert a large impact on both the magnitude and often the selectivity of neuronal responses.

Many neurons in visual areas and the posterior parietal cortex have such retinotopic receptive fields. They are typically much larger in the parietal cortex than in the primary visual areas. However, parietal neurons also carry information about eye position, as shown in a landmark experiment by Richard Andersen at the California Institute of Technology. He and his colleagues trained monkeys to fixate on a small cross on the projection screen while a parietal cortical neuron was recorded. After its receptive field had been mapped with a dot stimulus, the fixation point was varied to move the eyes in their sockets. The dot was also moved so that it always stayed in the center of the neuron's receptive field. Thus, the visual input to the preferred retinal spot remained the same throughout the experiment, while the eye position was systematically changed. The firing response of the parietal neuron varied as a function of the eye position. At some positions, the response was large, while at other positions it was smaller or even disappeared. This rate modulation had a spatial pattern, which the authors called the "gain field" (Figure 11.3). The firing rate of the neurons could be approximated by multiplying the firing rates induced by the receptive field profile with the factor due to modulation by eye position—hence the name "gain." This pioneering experiment revealed that the neuronal response to the retinal position of a visual object is multiplicatively scaled by eye position in parietal cortical neurons.²¹

Later experiments showed that visual responses of parietal neurons are also affected by head position and hand position. These eye and hand gain fields are not independent but instead have equal and opposite strengths, so that the eye–hand gain field combines reference frame transformations with movement planning at the same stage of cortical processing. Overall, these mechanisms in the parietal cortex guarantee that the direction of gaze is available for gain modulation regardless of whether the eyes or head are used to guide gaze.²²

21. These observations in area 7a of the monkey parietal cortex by Andersen et al. (1985) have been confirmed and extended to other cortical areas, including the ventral intraparietal area (VIP), by numerous other laboratories. Head position–modulated neurons are found in the lateral bank of the intraparietal sulcus (LIP; Brochier et al., 1995). Hand-position neurons are found in the medial bank, also called the *parietal reach region* (PRR), because PRR units are spatially selective for visual stimuli when they are used as the goal for reach movements (Chang et al., 2009).

22. In addition to eye, head, and hand positions, other types of signals also produce gain fields, including vergence of the pupils, target distance, chromatic contrast, and the expected probability of the sensory signals themselves, suggesting that gain modulation of neuronal activity is a general computational mechanism.

Retinal View and Movement Coordination

These neuronal mechanisms can explain how we reach for the volume knob on the radio while driving. As discussed in Chapter 10, parietal neurons are sensitive to the direction and speed of moving visual, auditory, and tactile stimuli, so this brain region is a good candidate to support this task. Through its connections to the premotor cortex, the parietal cortex can encode multimodal reference frames and perform coordinate transformations across reference frames. To perform the task correctly, parietal cortex neurons or their targets must calculate the relationship between the current gaze angle and the hand's target location, the knob on the radio.

In general, movement trajectory programming requires transforming information from sensory coordinates into a coordinate system appropriate for the eye, head, arms, or body. Instead of considering the entire population of parietal neurons, let's consider four, for simplicity (Figure 11.4). The goal is to guide the arm to the knob under visual control, regardless of the gaze angle.²³ Because the relationship between the body and the knob can be considered fixed for this simple situation, only the varying relationship between gaze and body must be calculated. As we have seen, parietal neurons “know” the position of the eyes in the orbit, and this information can be added to the object's retinal coordinates. By computing the summed activity of the four parietal neurons, the downstream reader neurons can infer the gaze–body relationship. The response magnitude of these reader neurons shifts as the gaze angle changes. Thus, the spatial coordinates of the arm's target (the knob) can be constantly updated as the gaze direction moves.

Unfortunately, this nice description is based on theoretical modeling rather than real population data and detailed physiological exploration of the transformation. Yet the principles by which gain-modulated neurons can transform coordinate systems illustrate a viable mechanism. Fortunately, some experiments support the existence of the hypothetical reader neurons of the coordinate transformation. In both parietal and premotor cortex, some neurons encode the horizontal direction and elevation of object locations independent of gaze direction. These neurons thus represent spatial locations in a head-centered frame of reference.²⁴

23. For this exercise, we disregard how the movement trajectory is organized. A large field of neuroscience is dealing with that complex issue (Shenoy et al., 2013).

24. Duhamel et al. (1997); Graziano et al. (1997). Fogassi et al. (1992) have shown that neurons in the premotor cortex of the monkey (inferior area 6) provide a stable body-centered frame of reference necessary for programming visually guided movements.

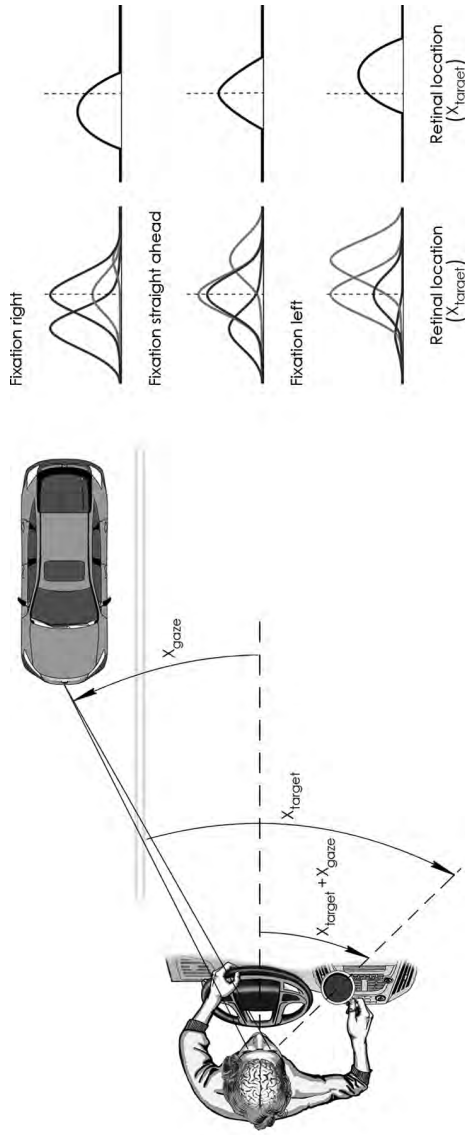


Figure 11.4. Gain fields support coordinate transformations. *Left:* A coordinate transformation performed by the visual system during driving. While driving, you may want to reach the volume control knob of the radio without shifting your gaze. The location of the knob relative to the body is given by the angle between the two dashed lines. For simplicity, assume that initially the hand is close to the body, at the origin of the coordinate system. The reaching movement should be generated in the direction of the knob regardless of where the driver is looking, that is, regardless of the gaze angle x_{gaze} . The location of the target in retinal coordinates (i.e., relative to the fixation point) is x_{target} , but this varies with gaze. However, the location relative to the body is given by $x_{\text{target}} + x_{\text{gaze}}$, which does not vary with gaze. By this mechanism, a change from retinal, or eye-centered, to body-centered coordinates is performed. *Right:* Combining the activity of several gain-modulated neurons may give rise to responses that shift. The left column shows the responses of four idealized, gain-modulated parietal neurons as functions of stimulus location; the three curves on the right are the summation of the four gain-modulated neurons, which correspond to the drive of a downstream observer unit. Inspired by Selinas and Sejnowski (2001).

Switching coordinate frames is a remarkable feat that involves discarding a lot of information and extracting only the essential features from a new point of view. The structures downstream from the visual-parietal areas need to know only what the eyes are looking at and where to reach, at least in our example. Reaching movements have to be learned by extensive motor practice, which provides the grounding for the transformed neuronal pattern.²⁵ Without action and feedback from its consequences, the neuronal responses to upstream inputs remain gibberish for the brain.

Abstraction by Different Reference Frames

It is remarkable that the transformation between eye-centered and body-centered coordinates requires only a neuronal gain mechanism. If such a simple mechanism works for one coordinate transformation, perhaps it can be deployed for similar purposes in other circuits. That is, the body-centered information carried by neurons that read the output from the parietal cortex can undergo further transformation at the next level of computation using the same type of gain control mechanism. Along the way, ever more complex features of the retinal information can be extracted.²⁶ When an object is viewed, approached, and touched from different directions, the multiple experiences may strip off the particular conditions in which the object was

25. While there is a backbone of anatomical connections between these layers, most information about the relationship among body parts, their distances from each other (Chapter 3), and their relations to the surrounding world have to be learned and maintained by practice. Experience can ground the sensory information by proprioceptive, corollary discharge signals by way of multiplicative gain modulation. The typical mistake of every novice car driver is steering the wheel in the direction that the head turns. This is the natural action for the brain. Commanding a big extracorporal extension, such as a car, is not rooted in our biology.

26. Salinas and Sejnowski (2001) and Salinas and Abbott (1995) were perhaps the first to hypothesize that gain modulation is a general-purpose coordinate transforming mechanism, extending early ideas by Zipser and Andersen (1988). Their modeling showed that neurons in a recurrently connected network with excitatory connections between similarly tuned neurons and inhibitory connections between differently tuned neurons can perform a product operation on additive synaptic inputs. Impairment of the gain control mechanisms in computer models, using the firing-rate gain-modulation mechanisms described earlier, can account for various deficits observed in hemi-neglect patients (Pouget and Sejnowski, 1997a). Gain control, thus, can be the cipher between different forms of representations.

sensed, leaving only its essential features, regardless of where they appear in the visual field. This is known as *translation invariance*, an abstraction.²⁷

GAIN CONTROL BY ATTENTION

The driving scenario can be mimicked in the laboratory by training monkeys to fixate their eyes to a cross (in lieu of traffic) and at the same time to pay attention to a particular location in their visual field where something may appear (in lieu of the volume knob). This is accomplished by extensive training, rewarded with a few drops of juice. The physiological measurement begins after an orientation-selective neuron is found and its receptive field identified in the visual-parietal system. Many neurons in these areas are called *orientation-selective* because they respond maximally to black-and-white stripes moving in a particular direction. By moving the striped gratings in multiple directions, the experimenter can construct a tuning curve. The measurements can be performed under two different conditions. In the first one, the monkey is trained to detect something in the receptive field of the neuron (to pay attention to it, as we do when we guide our hand to the radio knob by peripheral sight while keeping our gaze on the traffic ahead). In the second condition, the animal is asked to detect something in the opposite hemifield of vision (as we do when paying attention to the left mirror by peripheral vision). The two conditions are identical in every physical way because the monkey's gaze is fixed to a cross in the middle of the screen all the time, so only the location of attention is different. The shape of the two tuning curves is similar, as expected, but the magnitude of the response is larger when the monkey pays attention to the neuron's receptive field. The largest difference occurs at the peak of the receptive field; outside the receptive field, the two curves are equal. In short, selective attention multiplies or "gain-modulates" the neuron's response by a constant factor without changing its tuning characteristics. Attention-induced

27. Neurons in the inferior temporal (IT) cortex of the monkey (corresponding to the middle and inferior temporal gyri in the human brain) respond equally strongly to objects and, in some cases, to faces wherever they appear in the visual field and against whatever background. Face-selective neurons are more prevalent in the superior temporal sulcus. The same neurons may not respond at all when other objects or faces are shown at these locations (Desimone, 1991; Miyashita, 1993; Logothetis and Sheinberg, 1996). Neurons in the entorhinal cortex of human research participants show highly selective responses to objects and faces, such as the famous actress Jennifer Aniston. These neurons represent highly abstract features because even a cartoon or the written name of the actress was able to activate these cells (Quian Quiroga et al., 2005). This hypothetical coordinate transformation process is reminiscent of the formation of semantic information from episodic experience (Chapter 5).

gain modulation works well at all stages of the visual-posterior parietal cortical system²⁸ and likely in many other brain areas.

Autonomic Correlates of Attention

Attention is a spooky subjective term and an important item on James's list (Chapter 1). As a covert action, it is not accessible to direct investigation, short of placing numerous electrodes in the brain. But even if we remain motionless and our eyes do not rotate, there may still be some observable changes when we are concentrating on something. As discussed earlier, neuromodulators are important players in gain modulation. Two of these, acetylcholine and norepinephrine, also control the diameter of the pupils. When we are excited, norepinephrine levels are high, neuronal gain increases, and the pupils are constricted.²⁹ Thus, pupil diameter may be an objective readout of attention.

In an experiment addressing this possibility, human study participants were trained to recognize images with various visual and semantic features. Unknown to the participants, one visual feature and one semantic feature predicted monetary reward. Some participants were biased toward visual and others toward semantic features, which allowed the investigators to focus on the corresponding brain areas in the participants using functional imaging. Pupil response was strongly correlated with the degree to which learning performance followed these individual predispositions for visual or semantic features and also with the strength of the correlated activity across brain regions.³⁰ Thus, using the pupil diameter or other autonomic responses as a proxy for norepinephrine levels, neural gain can be studied in the intact human brain—at least indirectly. While there are obvious caveats to this approach, the findings nevertheless suggest that attention and gain modulation may have

28. Attention-induced gain modulation was discovered by Moran and Desimone (1985), recording from neurons in areas V1, V2, V4, and TEO, and replicated soon by work in area MT (Treue and Martinez-Trujillo, 1999) and in V2 and V4 (McAdams and Maunsell, 1999). The latter studies also showed that attention increases neuronal spiking multiplicatively by applying a fixed response gain factor. The review by Reynolds and Heeger (2009) discusses the many facets and advantages of attention-modulated gain control.

29. Pupil diameter is correlated with tonic levels of spiking activity in locus ceruleus noradrenergic neurons. Phasic activity of locus ceruleus neurons varies inversely with pupil diameter. Pupil diameter has been suggested to track attention and neural gain (Aston-Jones and Cohen, 2005; Gilzenrat et al., 2010; Reimer et al., 2016).

30. Eldar et al. (2013).

overt behavioral correlates.³¹ We can speculate that the output commands that control the pupil-affecting neurons send corollary discharges to parietal cortical regions which may be responsible for gain control of visual input.

GAIN CONTROL BY VELOCITY

Gain modulation of firing rates is also a prominent mechanism in the hippocampal-entorhinal system to accomplish running speed-independent place coding (Chapter 7). Each principal cell receives at least two types of inputs. One represents complex features arriving from various neocortical regions (external signals), and another represents information from the velocity system (internal signal). As a result, hippocampal and entorhinal neurons are under dual control. With two types of information at hand, one of which is action-related, the other external input can be grounded.

The firing rates of many place cells are elevated when the animal's locomotion speed increases (Figure 11.5).³² The middle of the place field shows the strongest effect, and it tapers off gradually toward the animal's entry and exit from the field. Outside the place field, velocity has no or very little impact on firing rate. As in other areas of the brain, the effect of the two inputs is multiplicative. Thus the place fields at different velocities show no change in field size but large changes in firing rates. Although the vestibular system is a likely source of speed modulation, it is not the only one. Firing rates are also modulated by speed during running on a wheel, on a treadmill, or on a ball

31. An obvious shortcoming is the lack of a distinction between global and focal attention, although the difference between visual and semantic feature-related changes in the blood oxygen level-dependent (BOLD) signal could be taken as an argument against a generalized arousal effect. Such global changes occur with stress, which is also associated with pupil contraction. Another important caveat is that gain control brought about by subcortical neuromodulators is inevitably slow due to the slow G-protein-coupled receptor responses, whereas gain modulation often occurs at a faster scale by other mechanisms. These fast mechanisms might also be at work during attention.

32. Each place cell has a relatively unique velocity rate profile. Some increase monotonically with rate; others increase and decrease their activity as a function of speed. The same neuron in two different place fields can have different speed rate profiles, indicating that current assembly members are driven by a unique distribution of speed inputs (McNaughton et al., 1983; Czurko et al., 1999). Not all place fields are modulated linearly with speed. Instead, a particular running speed is associated with maximum firing rates. A minority of place cells are negatively modulated by speed.

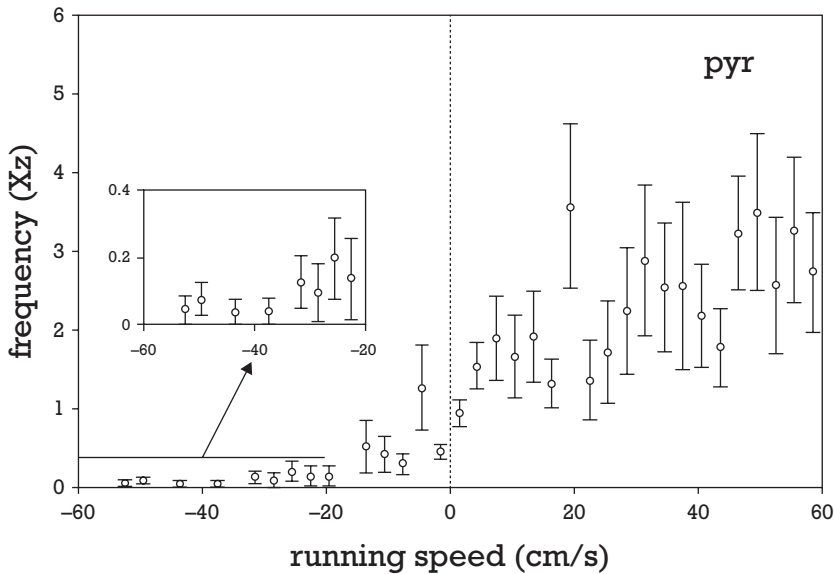


Figure 11.5. Running speed–dependent modulation of pyramidal cell (*pyr*) activity in the hippocampus. Negative and positive speed values indicate running directions (left or right) in a running wheel. Firing rate increases with speed when the rat faces the tuning field of the cell but is suppressed when it faces the opposite direction. The insert emphasizes speed-dependent suppression of firing. This tuning behavior is similar to the behavior of attention modulation in the receptive fields of visual cortical neurons.

or track with the head fixed, perhaps driven by optic flow or feedback from muscular activity.³³

The velocity input is essential for place cells to represent positions and distances reliably. Suppose that on a slow run the peak firing rate of a place cell is 10 spikes per second. During a faster run, the peak rate may increase to 15 spikes per second. In this case, the neuron fires 10 spikes per second when the animal moves away from the area of peak firing.³⁴ If neurons downstream from the hippocampus must decode the animal's location from firing rate alone, they will get different answers on different runs because 10 spikes per second corresponds to different positions depending on running speed. Even if the reader mechanisms get inputs from populations of place cells, the ambiguity of

33. Neither vestibular input nor optic flow is essential because speed affects the time shift across cell assemblies even in complete darkness while the head of the mouse is fixed (Villette et al., 2015).

34. A related solution is to use the ratios of instantaneous firing rates of place cells with overlapping place fields. In either case, a division is needed.

rate coding remains. To decode the animal's position correctly on the tract, the reader mechanism must divide the arriving pulses by 1.5 during the faster run.

This normalization solution is reminiscent of the mechanisms in intensity-invariant sensory neurons, which use both inhibitory and population feedback responses, as discussed at the beginning of this chapter. Therefore, one possibility is that, during fast runs, the rate increase of a particular place cell is normalized by the velocity-dependent increase of other neurons with overlapping place fields.³⁵ Another option is divisive inhibition. When the perisomatic inhibition of a place cell is selectively decreased experimentally, its firing rate is enhanced within the place field, with the highest change at the peak of the field. The decreased inhibition outside the place field has little effect. Because interneurons are also modulated by speed, this experimental manipulation shows that changes in inhibition constitute at least one of the factors that induce gain.³⁶

Velocity Affects Distance Representation

Velocity affects not only the firing rates of place neurons but their temporal relationships as well. You may recall from Chapter 7 that the cell assembly sequence starts during the late phase of the theta cycle with the representation of places the animal has already passed, and the spikes of subsequent assemblies advance to earlier phases representing upcoming locations. Each theta cycle, therefore, represents a sweep from past to the future or a segment of distance that starts behind and ends ahead of the animal. The forward-looking part of the representation is dubbed "look ahead." Thus, distances are converted into theta time scale durations. This distance-to-duration compression was discussed in Chapter 7.³⁷ To illustrate its significance, assume for a moment that the velocity signal plays no role in hippocampal computation. In that case, fixed time offsets between assemblies within the theta cycle should be related to variable run

35. In Chapter 7, we discussed that hippocampal place cells are velocity-controlled oscillators and that the theta phase of spikes advances proportionally to running velocity so that the theta phase of place cell spikes remains invariant to position. Thus, if the downstream readers have information about the phase of theta oscillation, a phase decoding mechanism is less ambiguous than rate decoding.

36. Royer et al. (2012). Several other experiments have shown that inhibition is, under most conditions, divisive rather than subtractive (Murphy and Miller, 2003; Carandini and Heeger, 2011; Wilson et al., 2012).

37. Skaggs et al. (1996); Tsodyks et al. (1996); Samsonovich and McNaughton (1997); Dragoi and Buzsáki (2006); Diba and Buzsáki (2007); Geisler et al. (2007); Johnson and Redish (2007).

durations between positions *A* and *B*. As a result, the distance representations would vary at different running speeds. The velocity gain is what prevents such distortion. Indeed, experiments show that velocity adjusts the transition times of place cell assemblies within the theta cycle. At faster velocities, the transition from one assembly to the next becomes proportionally faster. Because of the increased gain on assembly transitions, more cell assemblies can fit into a given theta cycle. In other words, larger segments of the environment are represented in each theta cycle during faster runs.

For this reason, distance “representation” (pardon me for using this convenient phrase) in the hippocampus does not vary much with the animal’s motor behavior.³⁸ Thanks to the velocity gain, we can judge the distances between objects irrespective of our speed; for example, when driving behind another car. This mechanism explains how the brain of a mounted archer can calculate the distance between himself and his target. We can conceive the velocity gain mechanism as an extension of the corollary discharge idea, with the goal of informing the brain that the changes in neuronal activity are brought about by the actions of the brain’s owner and not by changes in the environment.

FROM VELOCITY TO ATTENTION: INTERNALIZATION OF GAIN CONTROL

How can any of these mechanisms of gain control be mobilized by a subjective mechanism such as attention? During real-world navigation, the rate of change within assembly sequences is controlled by the animal’s body velocity or eye movements. As discussed earlier (Chapter 5), many environmentally driven functions become internalized as brains increase in complexity. For example, navigation becomes mental travel and serves seemingly different functions yet uses the same neuronal mechanisms. Self-organized assembly sequences in the hippocampal system and their disengagement from the external world seem perfectly suited for memory, imagination, and planning, but the internalized mechanism that controls gain during these functions is not clear.

38. Cell assembly sequence compression is measured as a ratio between distances of places (external measure by the experimenter) and the rate of change between the spikes of the place cells within the theta cycle (an internal mechanism). Maurer et al. (2012) calculated that sequence compression is high at low speed and decreases with the rat’s running velocity up to 40 cm/s. A potential explanation for why the compression factor reaches a plateau is because there should be enough time for the hippocampus to compute upcoming locations even at very high speeds.

We can hypothesize that attention is a candidate for this mechanism. In lieu of velocity, attention can adjust the rate of change of assembly sequences. This idea is supported by observations in the visual system. Investigators in several laboratories were surprised to observe that locomotion velocity exerts a robust effect on neuronal firing rate even in the primary visual cortex, showing a general effect of motion on neuronal gain.³⁹ The mediator of this speed-related gain is a special kind of inhibitory interneuron that inhibits other, dendrite-targeting inhibitory neurons.⁴⁰ The overall result is reduced inhibition of pyramidal cells, which increases their activity. These interneurons are excited by subcortical neuromodulators, such as the cholinergic input from the basal forebrain. This input increases its activity not only during locomotion but also when the animal attends to various stimuli. It remains to be explained how attention and cortical circuits can activate cholinergic neurons.⁴¹

If we tentatively accept that attention, as a multiplicative gain mechanism, can substitute for velocity, then we can explain how internalized functions of place cells, grid cells, and other types of neurons influence the dynamics of cognitive operations without locomotion. Even if the mechanisms are not clear at the moment, the advantage of this idea is that the evolutionary origin and behavioral meaning of attention gain in neuronal networks can be grounded by action-induced feedback to the brain.⁴² The vestibular system monitors the speed and acceleration of our own motion and can provide the ground truth about the rate of change of our actions. Although we do not know what neuronal mechanisms step in to substitute for vestibular signals, we do know that such a substitute exists because head-direction neurons maintain their relationships even during sleep, and their speed of change is controlled by internal brain dynamics (Chapter 5).

If division or multiplication underlying gain control and normalization have such far-reaching consequences on behavior and cognition, as we discussed

39. This was surprising only within the perceptual framework of vision, where action plays no role (Niell and Stryker, 2010).

40. This special interneuron type is known as the *vasoactive intestine peptide* (VIP)-expressing subgroup, which can be activated by both acetylcholine via nicotinic receptors and serotonin via 5-HT_A receptors (Rudy et al., 2011; Fu et al., 2014). This VIP neuron-mediated disinhibitory circuit may constitute the common gain control mechanism activated by both locomotion and attention.

41. Gielow and Zaborszky (2017); Schmitz and Duncan (2018).

42. Krauzlis et al. (2014) also argue that attention is an effect rather than a causal agent. It is a byproduct of circuits centered on the basal ganglia involved in estimating the current state of the animal and its environment. Brain mechanisms responsible for attention existed prior to the emergence of the neocortex.

earlier, one might expect that they exert a profound effect at multiple levels of neuronal organization. And they do. The next chapter discusses how.

SUMMARY

Gain and normalization are simple but fundamental mechanisms that can support numerous functions in the brain. These mechanisms are called by a variety of names, such as coordinate transformation, place anchoring, abstraction, and attention. They are of fundamental importance, as illustrated by the numerous mechanisms for gain control that exist in the brain, including divisive inhibition, short-term plasticity of synapses, and subcortical neuromodulation. Gain control allows inputs from the retina and the positions of the eyes in their sockets, the head, and the hands to affect the magnitude of responses to visual inputs in multiple brain regions, particularly the parietal cortex. Gain control mechanisms can shift coordinate representations, for example, from visual space to head space to hand space or recognize an object as the same when it is viewed from different directions. The mechanisms of translation and object invariance are the neuronal basis of abstraction, a process of ignoring features that are not essential to recognizing entities. Gain control is important in the hippocampal system, allowing judgment of distances independent of locomotion speed. Attention may be viewed as internalized gain control.